

PaySight

UPI Reliability & Fraud Analytics

Identifying Transaction Failure Patterns and Fraud Risk Using Data Analytics

Project Problem Statement • Individual Data Analytics Project • 2026

1. Background

UPI and other digital payments handle huge transaction volumes every day. Two problems keep coming up for any payments team: why transactions fail, and how to catch fraud before it costs money and trust.

Failures happen for simple reasons network issues, low balance, system errors during busy hours. Fraud is trickier fraud patterns change fast, so old rule-based checks stop working. This project looks at both using real data instead of guesswork.

2. Business Problem

The way a fintech payments team would see this problem:

- Transaction failure rates are rising and there's no clear breakdown of root causes by type, time, or channel.
- Fraudulent transactions cause direct financial loss and are hard to distinguish from genuine high-value transfers.
- There's no single view of which transaction patterns are high-risk versus which are simply high-value.
- Leadership lacks a live dashboard to track success rate, fraud rate, and financial exposure day to day.

Goal: give management real numbers on why transactions fail and where fraud happens — not guesses.

3. Project Objective

Build one pipeline, start to finish raw data to a working dashboard using SQL, Python, statistics, and Power BI, plus a simple ML model for fraud scoring:

1. Clean and validate raw transaction data so every downstream number is trustworthy.
2. Store and query the data in SQL to compute business metrics reliably.
3. Use Python for exploratory analysis and statistical testing of failure/fraud drivers.
4. Build a basic ML model (logistic regression / decision tree) to score fraud risk per transaction.
5. Design a Power BI dashboard that gives leadership a real-time view of reliability and fraud KPIs.
6. Document every step so the project reads as production-style work, not a one-off notebook.

4. Dataset

Dataset: PaySim — Synthetic Mobile Money Transaction Dataset

Source: Kaggle — <https://www.kaggle.com/datasets/ealaxi/paysim1>

Size: ~6.3 million transactions over 30 simulated days. Real UPI data is never public, so this is the closest available stand-in — it's built from real mobile-money transaction patterns.

Data dictionary:

Column	Description
step	Time unit (1 step = 1 hour) since simulation start
type	Transaction type: PAYMENT, TRANSFER, CASH_OUT, CASH_IN, DEBIT
amount	Transaction amount (local currency)
nameOrig	Customer ID initiating the transaction
oldbalanceOrg	Sender balance before transaction
newbalanceOrig	Sender balance after transaction
nameDest	Recipient customer/merchant ID
oldbalanceDest	Recipient balance before transaction
newbalanceDest	Recipient balance after transaction
isFraud	Ground-truth fraud label (1 = fraudulent)
isFlaggedFraud	Flagged by the simulator's rule-based control (large transfer heuristic)

5. Approach

5.1 Data Cleaning (Python + Excel)

- Check for missing data, duplicate rows, and balances that don't make sense.
- Fix column types — amounts as numbers, dates as dates.
- Flag transactions where the balance math doesn't add up — useful data-quality and fraud signal.
- Save one final cleaned file, locked with a checksum. SQL and Power BI both read from this one file — fixes the 'numbers don't match' problem.

5.2 SQL Layer

- Load the cleaned data into a database (PostgreSQL/MySQL).
- Write queries for the real business questions — failure rate by type, fraud by amount, fraud by time, risky accounts.
- Run the same checks in SQL that Python already ran — row count, total amount, fraud count — to confirm nothing changed in between.

5.3 Statistics (Python)

- Test whether amount, type, and time actually affect fraud, or if it just looks that way.
- Check if the balance-mismatch flag is actually linked to real fraud.
- Look at amount distributions to set sensible risk brackets.

5.4 Basic Machine Learning

- Simple model (logistic regression / decision tree) to score fraud risk per transaction — not production-grade, just enough for a working risk score.
- Judge it on precision/recall, not accuracy — fraud is rare, so accuracy alone is misleading.

5.5 Power BI Dashboard

- Dashboard connects to the SQL database directly, not the raw CSV — numbers always match.
- Pages: Executive Summary, Failure Analysis, Fraud Analysis, Risk Scoring, Trends Over Time.
- PaySight color theme: deep blue for brand, teal for good/reliable numbers, red only for fraud/failure alerts, light grey background.

6. Business Questions This Project Answers

7. Which transaction types have the highest failure rate?
8. Which transactions are most likely to become fraudulent?
9. How do transaction amounts influence fraud probability?
10. Which time periods experience the highest transaction failures?
11. What customer behaviors indicate suspicious activity?
12. Which operational factors reduce payment success rates?
13. How can the company improve transaction reliability?
14. What business actions can reduce fraud losses?

7. Expected Business Outcomes

- Clear view of the top drivers behind transaction failures.
- Early detection signals for suspicious transaction patterns.
- A live KPI dashboard for payment success rate and fraud exposure.
- A documented, reproducible pipeline where Python, SQL, and Power BI numbers always agree.
- A portfolio-ready case study demonstrating end-to-end fintech analytics skills.

8. Tools & Technologies

Tool	Purpose
SQL (PostgreSQL/MySQL)	Data storage, business queries, validation cross-checks
Python (Pandas, NumPy, SciPy, Scikit-learn)	Data cleaning, EDA, statistical testing, basic ML fraud scoring
Power BI	Interactive executive dashboard & reporting
Excel	Initial spot-checks and data validation
GitHub	Version control and project documentation

9. Deliverables

- Data cleaning & validation report
- SQL database with business + validation queries
- Python EDA notebook and statistical analysis report
- Basic ML fraud-risk scoring model
- Power BI dashboard (PaySight theme)
- Business insights report answering all 8 business questions
- Executive presentation deck
- GitHub repository with full documentation

10. Success Criteria

- Primary drivers of transaction failure clearly identified and quantified.
- Fraud-related patterns detected and explained with supporting statistics.
- Every number in the dashboard traceable back to the same validated source — no mismatches between Python, SQL, and Power BI.
- Recommendations are specific and backed by data, not generic statements.
- End result is professional enough to present as a fintech-industry portfolio case study.